

Solution Requirements

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (disaster management officials or administrators) can securely access the application using login credentials. (If your project has no login, you can write: "No authentication is required as the application is intended for public access.")	Medium
2	Authorization levels	Administrators can manage the system and prediction records, while users can only enter weather parameters and view prediction results.	Medium
3	External interfaces	The system accepts weather data from the Kaggle dataset and can optionally integrate with Weather APIs (e.g., IMD/OpenWeather) and IBM Cloud services.	High

4	Transactions processing	The system validates user inputs, preprocesses weather parameters, sends them to the trained ML model, and returns the flood prediction result.	High
5	Reporting	The application displays flood prediction results and can generate prediction history and performance reports for analysis.	High
6	Business rules, process etc.	Predictions are generated only after all mandatory weather parameters are entered. The model classifies the flood risk based on the trained machine learning algorithm.	High
7	Compliance to laws or regulations	The system stores only non-sensitive environmental data and follows basic data privacy and ethical AI principles.	Medium
8	Other	The system allows retraining of the ML model using updated flood datasets to improve prediction accuracy over time.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate flood predictions quickly after the user submits weather data.	Prediction response time < 3 seconds
2	Scalability	The application should support multiple users and larger datasets without significant performance degradation.	Supports 500+ concurrent users
3	Security & Data Privacy	User inputs and prediction data should be securely processed and protected from unauthorized access.	Secure communication using HTTPS and controlled access
4	Reliability & Availability	The application should remain available during normal operation with minimal downtime.	99% uptime during deployment

5	Usability & Accessibility	The application should provide a simple, intuitive interface that can be easily used by disaster management personnel.	Easy navigation with responsive web interface
6	Other	The system should be easy to maintain, update with new datasets, and deploy on different platforms including IBM Cloud.	Modular code and successful deployment on IBM Cloud / Local Server